Ch. 7: Data Analysis, 2017 data, pt. 3 - Total Intensity by Total Area

Area and Intensity vary in tight relation to each other.

Import Libraries

In [1]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib as mpl
from matplotlib import pyplot as plt

%matplotlib inline
```

Import the data

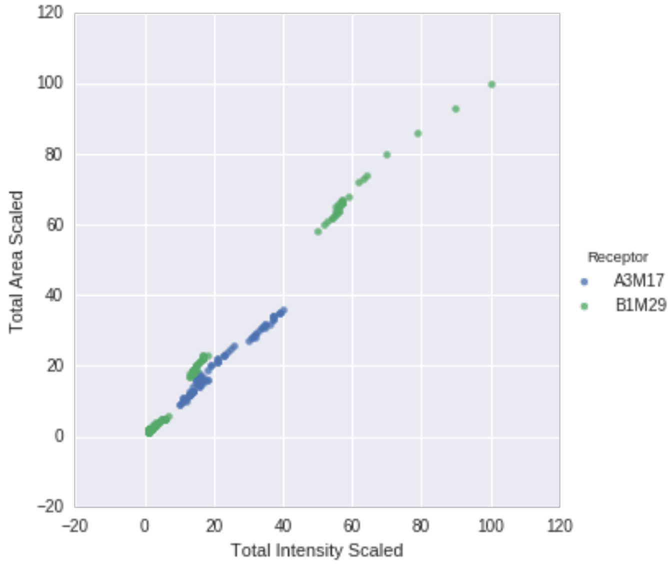
In [2]:
```
totalsXX = None
totalsXX = pd.read_csv('TXX_Totals.csv')
```

Visualizations

Unless otherwise noted, the graphs will use the scaled values.

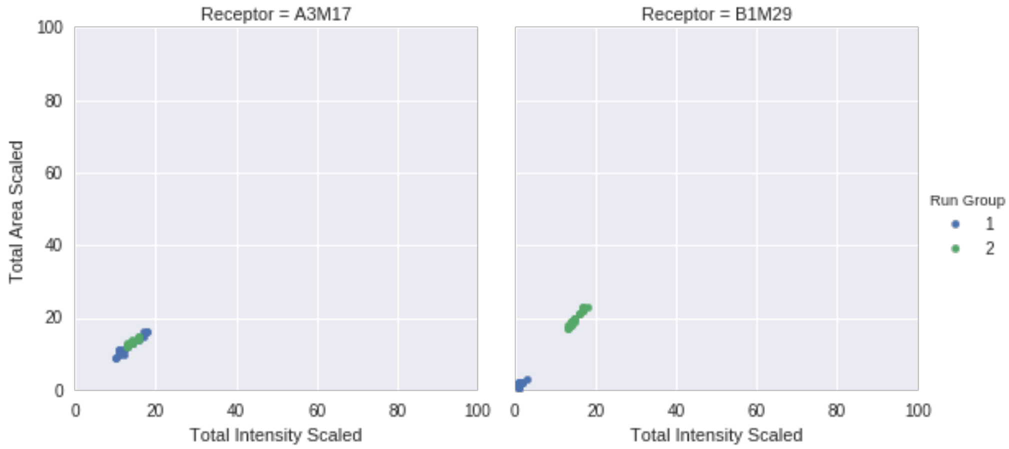
Intensity by area, both receptors, post steps, both runs

In [3]:
```
intensity_by_area_all = sns.lmplot('Total Intensity Scaled', 'Total Area Sca
led',
                                   data=totalsXX,
                                   fit_reg=False,
                                   hue='Receptor')
intensity_by_area_all.savefig('xx_intensity_by_area_both.png')
```



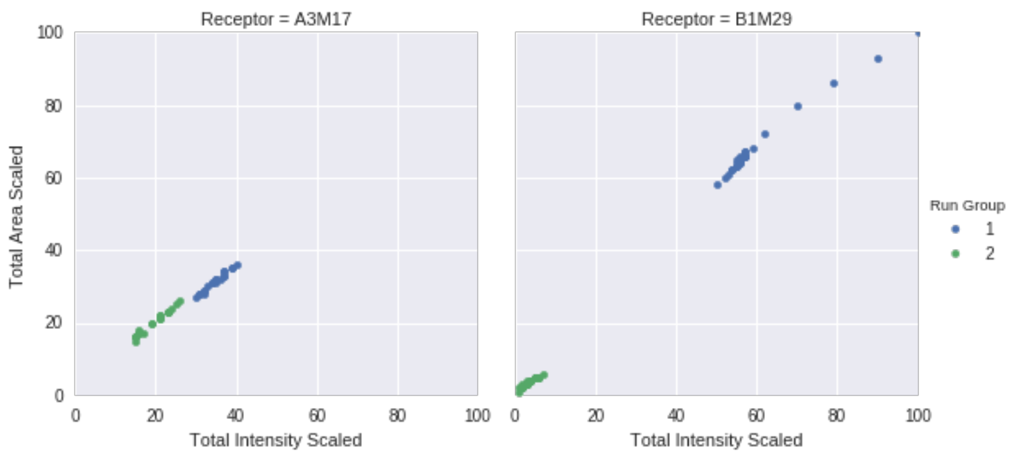
Intensity by area, both receptors, -nl-post

In [4]:
```
tmpdf = totalsXX.loc[totalsXX['Experiment Step'] == '-nl-post']

g = sns.FacetGrid(tmpdf,
                  col='Receptor',
                  col_order=['A3M17', 'B1M29'],
                  col_wrap=2,
                  hue='Run Group',
                  size=4,
                  ylim=(0,100),
                  xlim=(0,100)
                 )
g.map(plt.scatter, 'Total Intensity Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('xx_intensity-by-area_both-receptors_noligand-post.png')
```



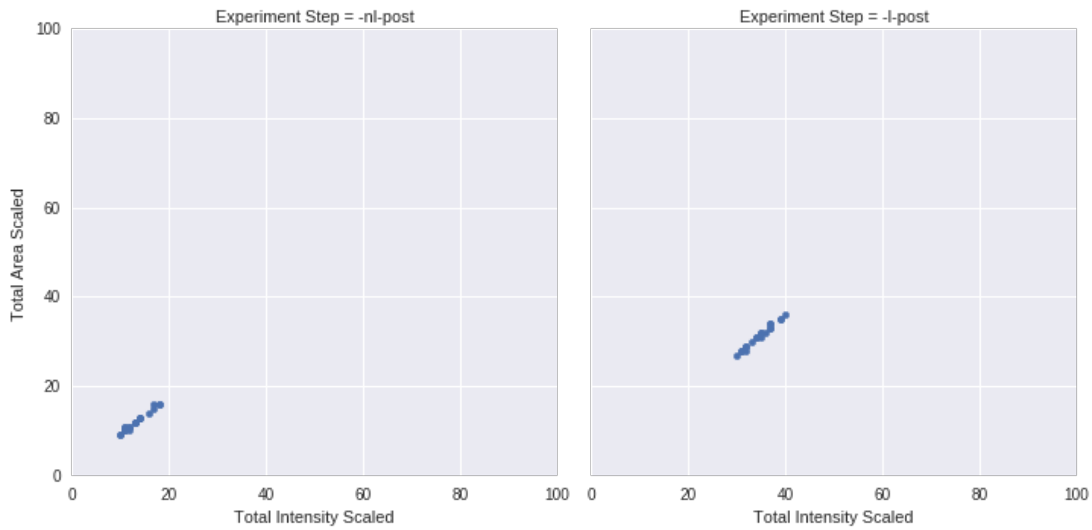
Intensity by area, both receptors, -l-post

In [5]:
```
tmpdf = totalsXX.loc[totalsXX['Experiment Step'] == '-l-post']

g = sns.FacetGrid(tmpdf,
                  col='Receptor',
                  col_order=['A3M17', 'B1M29'],
                  hue='Run Group',
                  col_wrap=2,
                  size=4,
                  ylim=(0,100),
                  xlim=(0,100)
                  )
g.map(plt.scatter, 'Total Intensity Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('xx_intensity-by-area_both-receptors_ligand-post.png')
```



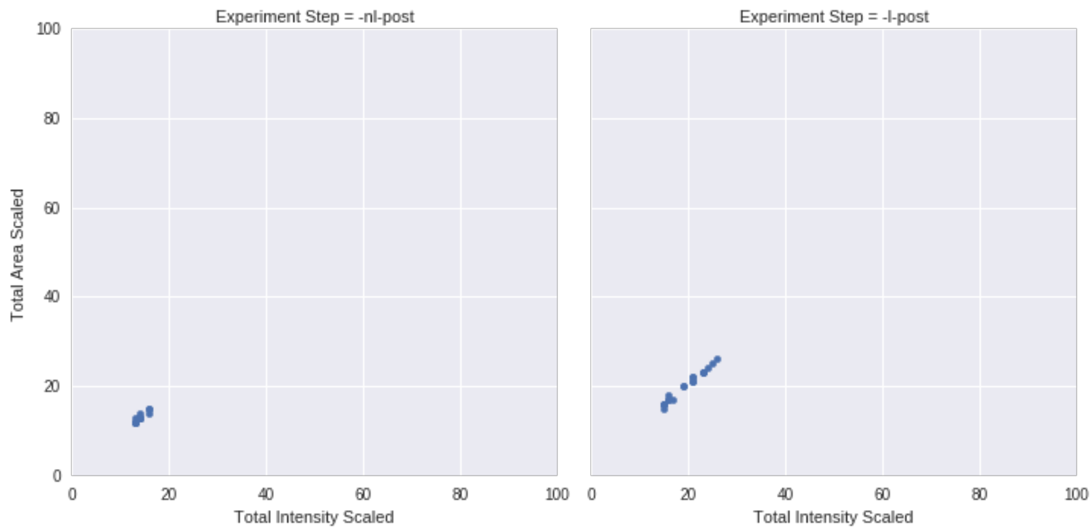
Intensity by area, A3M17, post steps, run group 1

In [6]:
```
tmpdf = None
tmpdf = totalsXX.loc[(totalsXX['Receptor'] == 'A3M17') &
                     (totalsXX['Experiment Step'].str.contains('post')) &
                     (totalsXX['Run Group'] == 1)
                    ]

g = sns.FacetGrid(tmpdf,
                  col='Experiment Step',
                  col_order=['-nl-post','-l-post'],
                  col_wrap=2,
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Intensity Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('xx_intensity-by-area_a3m17_post-steps_run-group-1.png')
```



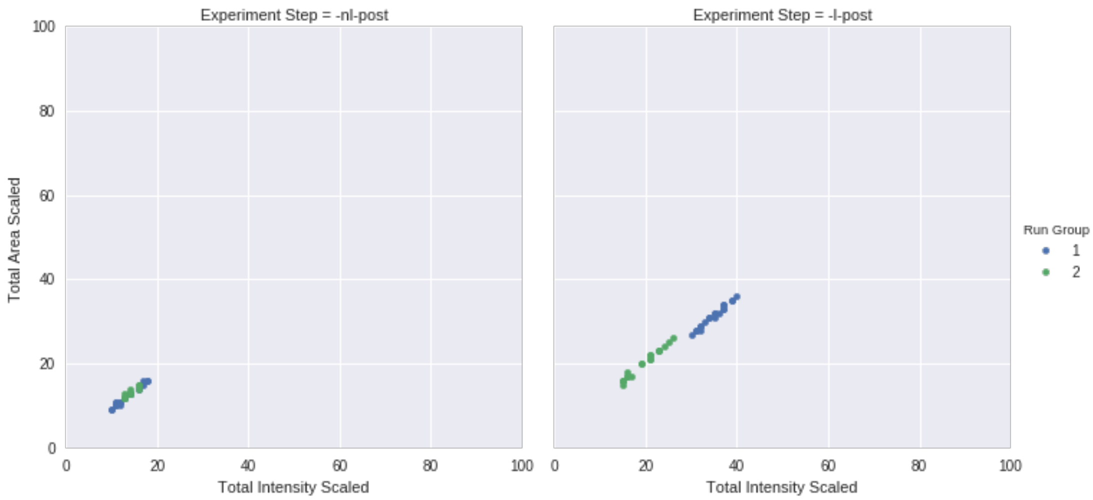
Intensity by area, A3M17, post steps, run group 2

In [7]:
```
tmpdf = None
tmpdf = totalsXX.loc[(totalsXX['Receptor'] == 'A3M17') &
                     (totalsXX['Experiment Step'].str.contains('post')) &
                     (totalsXX['Run Group'] == 2)
                    ]

g = sns.FacetGrid(tmpdf,
                  col='Experiment Step',
                  col_order=['-nl-post','-l-post'],
                  col_wrap=2,
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Intensity Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('xx_intensity-by-area_a3m17_post-steps_run-group-2.png')
```



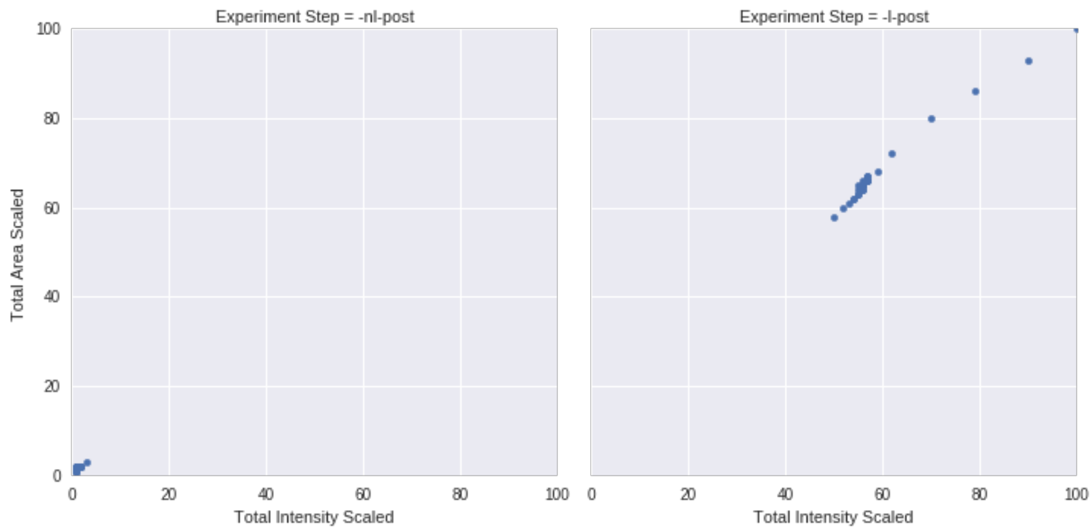
Intensity by Area, A3M17, post steps, both run groups

In [8]:
```
tmpdf = None
tmpdf = totalsXX.loc[(totalsXX['Receptor'] == 'A3M17') &
                     (totalsXX['Experiment Step'].str.contains('post'))
                    ]

g = sns.FacetGrid(tmpdf,
                  col='Experiment Step',
                  col_order=['-nl-post','-l-post'],
                  hue='Run Group',
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Intensity Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('xx_intensity-by-area_a3m17_post-steps_both-run-groups.png')
```



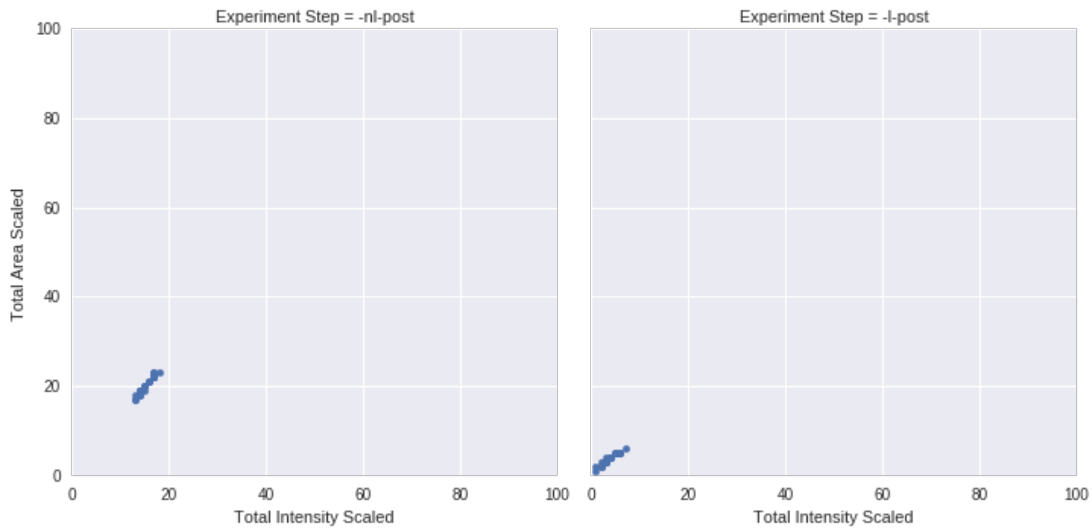
Intensity by area, B1M29, post steps, run group 1

In [9]:
```
tmpdf = None
tmpdf = totalsXX.loc[(totalsXX['Receptor'] == 'B1M29') &
                     (totalsXX['Experiment Step'].str.contains('post')) &
                     (totalsXX['Run Group'] == 1)
                    ]

g = sns.FacetGrid(tmpdf,
                  col='Experiment Step',
                  col_order=['-nl-post','-l-post'],
                  col_wrap=2,
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Intensity Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('xx_intensity-by-area_b1m29_post-steps_run-group-1.png')
```



intensity by area, B1M29, post steps, run group 2

In [10]:
```
tmpdf = None
tmpdf = totalsXX.loc[(totalsXX['Receptor'] == 'B1M29') &
                     (totalsXX['Experiment Step'].str.contains('post')) &
                     (totalsXX['Run Group'] == 2)
                    ]

g = sns.FacetGrid(tmpdf,
                  col='Experiment Step',
                  col_order=['-nl-post','-l-post'],
                  col_wrap=2,
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Intensity Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('xx_intensity-by-area_b1m29_post-steps_run-group-2.png')
```



Intensity by Area, B1M29, post steps, both run groups

In [11]:
```
tmpdf = None
tmpdf = totalsXX.loc[(totalsXX['Receptor'] == 'B1M29') &
                     (totalsXX['Experiment Step'].str.contains('post'))
                    ]

g = sns.FacetGrid(tmpdf,
                  col='Experiment Step',
                  col_order=['-nl-post','-l-post'],
                  hue='Run Group',
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1)

g.map(plt.scatter, 'Total Intensity Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('xx_intensity-by-area_b1m29_post-steps_both-run-groups.png')
```